

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
 18 emerging paradigms in neuroscience—specifically classical electrody-
 19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 20 tion, culminating in a topological fixed point: a unified “Self.” We
 21 state the claim narrowly, because the subject matter invites a wider
 22 reading than the work supports: no new physics is derived here. Every
 23 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 24 ble’s classification of topological defects, Landauer’s erasure bound,
 25 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-
 26 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed
 27 point. What is new is that they are stated in one formal language,
 28 their consistency mechanically checked, and the points where the chain
 29 relies on stipulation rather than derivation located exactly. The de-
 30 velopment declares no axioms and every structure carrying physical
 31 content is inhabited by a witness; the synchronization step is proved
 32 from a dynamics for initial data inside an explicit basin and assumed
 33 outside it, and the identification of a global section of the probability
 34 sheaf with the unity of experience is a stipulation, marked as such
 35 throughout rather than absorbed into the derivation. The step from
 36 a coarse-grained coupling kernel to the brain’s endogenous electro-
 37 magnetic field is likewise an empirical commitment and not a deduc-
 38 tion; we separate it from the coarse-graining theorem that precedes it,
 39 weaken it to a modulatory-but-integrative role, and give four condi-
 40 tions that would falsify it. The framework is nonetheless falsifiable in

one place that is its own and no rival's: recovery from sudden awakening should follow a delayed sigmoid rather than an exponential, because the variable gating it is astrocytic volume on a timescale of seconds to minutes rather than cortical phase realignment on a timescale of milliseconds. As a corollary, we show why rigid von Neumann architectures (e.g., GPUs) are thermodynamically disadvantaged or structurally constrained from reaching these specific continuous thermodynamic attractors.

1 Introduction

The pursuit of a physical theory of consciousness typically involves searching for the biological correlates of mind or treating subjective experience as an abstract computational algorithm. This paper asks a more fundamental question: If we assume only basic physical primitives—continuous symmetries (the Poincaré group), geometry, and thermodynamics—is the emergence of a unified, perceiving entity a natural physical attractor?

We propose a framework where the structural properties of consciousness are derived as physical attractor states. Establishing a deductive chain from basic principles, we demonstrate how continuous symmetries break to form boundaries, how those boundaries dissipate free energy as non-equilibrium steady states, and how their evolution along paths of least action results in physical resonance with their environment. This eventually scales into the macroscopic electromagnetic unity proposed by emerging field theories

63 of mind. To avoid critiques of “math theatre,” we delineate between literal
 64 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 65 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 66 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 67 spective aligned with Russellian Monism: objective measurements of macro-
 68 scopic electromagnetic fields constitute the third-person “outside,” while the
 69 system’s internal physical deformation to reach structural resonance consti-
 70 tutes the first-person intrinsic experience.

71 **What is new here, and what is not**

72 A framework paper about consciousness invites a particular suspicion, and it
 73 is better answered at the start than at the end. **No new physics is derived**
 74 **here.** Every physical result the chain invokes is established, and most of
 75 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 76 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 77 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 78 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
 79 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
 80 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy
 81 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 82 Bell and Crooks relating dissipation to nonpredictive information (Still et
 83 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 84 dissipated work with the relative entropy between forward and time-reversed

85 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
86 1922). A physicist reading this paper will recognize all of it and would
87 establish the point in three paragraphs if we did not. Conceding it costs
88 nothing.

89 What is new is of a different kind, and it is a smaller claim than a theory
90 of consciousness usually makes. These results have been stated in one formal
91 language, their mutual consistency has been mechanically checked, and the
92 places where the chain relies on stipulation rather than derivation have been
93 located exactly. The value of a machine-checked chain is not that any link is
94 novel; it is that no link can quietly borrow from another, that a hypothesis
95 cannot be smuggled in as a definition, and that the seams — where physics
96 stops and interpretation begins — are fixed in the source rather than nego-
97 tiable in the prose. Section 2.1 reports what the check cost: three of five
98 original axioms each proved **False**, from one shared root cause, and none of
99 the three had been caught by reading.

100 The framework is not thereby unfalsifiable, and one of its predictions is
101 its own rather than a rival’s. Section 11 argues that recovery from sudden
102 awakening should follow a *delayed sigmoid* rather than an exponential, be-
103 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
104 clearance, on a timescale of seconds to minutes, while the cortical rhythms
105 whose realignment is usually taken to constitute the recovery move in mil-
106 liseconds. That is a claim about functional form; it is fittable against data of
107 a kind already collected; and a purely electrical account of the same recovery

108 predicts a different form.

109 **2 Formalization in Lean 4**

110 A core contribution of this work is the verification of our deductive chain
111 using the Lean 4 interactive theorem prover. This approach builds on recent
112 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
113 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
114 2026). It is necessary to explicitly distinguish which claims have been math-
115 ematically derived as theorems within the proof assistant, and which remain
116 irreducible physical postulates. Figure 1 shows the chain with the status of
117 every link marked, and Table 1 states in five rows what each link assumes.
118 The complete claim-by-claim map to Lean identifiers, which runs to several
119 pages, is Table S1 in the supplement; the main text does not depend on
120 reading it.

121 **2.1 Soundness of the formalization**

122 Formalization is only as trustworthy as the axioms it rests on, and an axiom
123 that is inconsistent silently makes every theorem downstream of it vacuous
124 while leaving the build green. We report a case of exactly this, found by
125 auditing our own development, because the failure mode is easy to reproduce
126 and we have not seen it discussed in the applied-formalization literature.

127 Earlier drafts of this work declared five axioms. Three of them were



Figure 1: The deductive chain, with the status of each link marked. ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ vocabulary, consumed by no theorem. The two weak links are marked rather than argued around: the identification of the coarse-grained kernel with the cortical electromagnetic field is empirical, and the final step to experience is stipulated. Table 1 states what each link assumes; the full claim-by-claim map to Lean identifiers is Table S1 in the supplement.

Claim	Status	What is assumed
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasure costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
Discrete coupling coarse-grains to a continuous field; that field is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, glues into a global section, and carries a unique reflexive fixed point	Theorem	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated; contraction on a complete metric space. The step from fixed point to experience is stipulated.

Table 1: The chain in five claims. “Class field” means a physical postulate that each model discharges for itself rather than a global axiom; the development declares **no axioms**, and `#print axioms` on any result reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent (Section 2.1). The complete map from claims to Lean identifiers, with the scope limitation of each, is Table S1 in the supplement.

128 individually inconsistent: each, on its own, proves `False`. All three shared
 129 one root cause — an axiom that constrains a symbol it does not itself bind.
 130 Concretely:

- 131 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
 132 energy functional as an *implicit* argument occurring only in its hypoth-
 133 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
 134 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
 135 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
 136 this proves $1 = 0$.
- 137 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
 138 a bath-entropy difference *for every instance* of the class. An instance
 139 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- 140 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ
 141 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
 142 point mass against a uniform distribution refutes it.

143 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
 144 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`
 145 — and are refutable as soon as the probability presheaf admits two distinct
 146 global sections. (Both were demoted to class fields at the time; both have
 147 since been removed from the class altogether, for a separate reason given in
 148 Derivation 5.)

149 The correction is uniform, and we state it as a rule we now follow: *a*
 150 *physical postulate that mentions a class field must be a field of that class,*
 151 *never a standalone axiom quantified over all instances.* A postulate should
 152 be an obligation each model discharges, not a global claim about every model.
 153 Recast this way, the development declares no axioms at all. Two of the five
 154 became genuine theorems — pointwise vacuum minimization is now proved
 155 from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs' inequality was already one
 156 — and the other three became instance obligations.

157 This makes non-vacuity the live question, since an uninhabitable class
 158 makes its theorems vacuous just as surely as an inconsistent axiom did.
 159 **Examples.lean** therefore constructs witnesses: a one-bit erasure system with
 160 a genuine bath that discharges **heat_eq** and reproduces $\Delta Q = k_B T \ln 2$;
 161 a perfectly resonant system discharging **kl_bound**; and a scalar field on
 162 a one-point spacetime discharging **ActionPrinciples**. Derivation 5's glu-
 163 ing structures are witnessed too, which was until recently the largest open
 164 gap: **Examples.lean** builds a **ThermodynamicCover** over a three-site sub-
 165 strate carrying the discrete topology and its Borel σ -algebra, covered by two
 166 *overlapping* open patches whose intersection is a single shared site. The
 167 local sections are germ families of a phase-indexed finite measure $\mu_\theta =$
 168 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 169 family is provably non-constant in θ (**phaseMeasure_not_const**), so the pe-
 170 riodicity obligation of the constructor **LocalSectionSynchronization.of-**
 171 **InvariantMeasure** is discharged by an argument about $\cos$ rather than satis-

172 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 173 rather than assumed. `global_section_from_thermodynamics` then fires on
 174 this instance and glues the two patch-local sections into a unique global one.
 175 Two further covers answer the follow-up question of how non-degenerate such
 176 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 177 ing: the phase field can be made non-constant as a function, but not as a
 178 physical state, because `thermodynamic_equilibrium` forces every instance
 179 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 180 round, from two patches with independently chosen mass profiles that meet
 181 only on the site they share, and the section the gluing returns is neither of
 182 them. That is the same kind of result as `contracting_implies_const` in
 183 Derivation 6 — a proof that the witness is as strong as the class permits,
 184 rather than a witness that could be improved by trying harder.

185 What the witnesses establish is that the classes are inhabitable, so the
 186 theorems about them are not vacuous. What they do not establish is that
 187 a cortex is such an instance. In the cover above the phase field is already
 188 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 189 the equilibrium obligation dischargeable; getting a physical system into that
 190 configuration is the content of the informal dynamical argument, not of the
 191 Lean development.

192 The same audit turned up a defect of a different kind, worth record-
 193 ing because it is not an axiom problem and would not have been caught
 194 by `#print axioms.mesh_refinement_convergence` — the claim that dis-

195 create edge weights converge to the continuous energy as the mesh is refined
 196 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
 197 not merely unproved but false. Nothing required the edge regions to cover
 198 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
 199 all empty satisfied the mesh hypothesis while contributing nothing to the
 200 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
 201 quantified over a triangulation and then resolved the vertex type by instance
 202 search rather than through the bound variable, so the variable was never
 203 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
 204 gence is now stated along a *sequence* of triangulations of a named region
 205 S , each required to be regular — edge regions measurable, without self-
 206 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 207 plicial data in the sense that a vertex lies in the closure of every region
 208 of an edge at it. Closure rather than membership is forced: cells are dis-
 209 joint, so a shared vertex belongs to at most one of them. The manuscript’s
 210 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 211 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 212 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 213 tion, reading the double sum as a Riemann sum would be a category error.
 214 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
 215 the integrand, and convergence follows for uniformly continuous integrands
 216 as the mesh shrinks. The regularity conditions pull against each other, so
 217 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into

218 half-open cells and checks that the approximation genuinely moves — $\frac{1}{2}$ at
 219 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 220 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 221 the accompanying simulation measures; that would need a midpoint error
 222 term with a second derivative, which we have not developed.

223 A third defect was one of bookkeeping across files, and it is the kind that
 224 survives any number of green builds because each half is individually cor-
 225 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 226 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 227 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 228 — phase-locked in both directions — and everything downstream of it, in-
 229 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 230 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 231 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 232 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 233 it has no minimum for phase-locking to attain, and the sentence “the phase-
 234 locked state minimizes the Lyapunov potential” was, read literally, false of
 235 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 236 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 237 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 238 frequency system — which is exactly the system Derivation 5 already worked
 239 with — the two potentials coincide there, and the descent result transports
 240 onto the coupling potential whose minimizers are the phase-locked states.

241 Phase differences, phase-locking and the magnitude of the order parameter
 242 are all proved invariant under the change of frame, so nothing observable
 243 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 244 two oscillators with a common frequency, discharging every hypothesis. The
 245 restriction is real and we state it plainly: the reduction is exact only for
 246 identical frequencies. A genuine spread of natural frequencies leaves residual
 247 detunings in the reduced system, phase-locking then requires coupling above
 248 a critical K_c , and that K_c — Kuramoto’s threshold for a spread of frequen-
 249 cies, not the noise threshold $2D$ of Section 10 — is nowhere in our Lean
 250 development. It is studied numerically, in Figure 5, as Table S1 records.

251 A fourth defect was dimensional, and it concerned the one row of Table S1
 252 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 253 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 254 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 255 against an unnormalized measure carries the substrate’s mass twice over,
 256 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 257 The comparison therefore depended on how large the substrate was rather
 258 than on how strongly it was coupled, and any kernel whatever could be
 259 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 260 ure concretely: on a two-site substrate carrying counting measure, a sys-
 261 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 262 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 263 satisfied the old predicate. The fix is to require the substrate’s measure to

264 be a probability measure, which is the normalization in which the mean-
 265 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 266 then gives that a constant kernel has mean-field strength exactly K , and
 267 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 268 We flag what this does and does not do. It makes the predicate say something
 269 whose truth depends only on the physics, and §9 witnesses it on both sides of
 270 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 271 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 272 at $K = 2D$, which is a separate question from whether the predicate is di-
 273 mensionally sound. That bifurcation is proved below, in both directions,
 274 from an assumed stationary density; deriving the density itself needs the
 275 Fokker–Planck operator and is not done.

276 We draw no grand moral from this beyond the obvious one. The value
 277 of a proof assistant here was not that it certified our argument, but that it
 278 made a specific, checkable claim about what had been assumed — and that
 279 claim turned out to be false in a way no amount of reading the prose would
 280 have revealed.

281 **3 Axiom 1: Symmetries and Phase Space**

282 Our starting premise relies on fundamental geometry. In our Lean formaliza-
 283 tion (`Phase1_Primitives.lean`), continuous spacetime symmetries (e.g., the
 284 Poincaré group) act upon field states, and the invariance of the system’s total

285 energy under that action is recorded by the class `SymmetryInvariantAction`.
 286 Physically, it is such an invariance that Noether’s theorem converts into the
 287 conservation of energy and momentum. The scope of the *formal* claim is nar-
 288 rower and we state it plainly: the invariance is a class field, i.e. an assumption
 289 a model must discharge, and no conserved quantity is ever constructed from
 290 it. There is no Noether theorem in our Lean, and nothing downstream of this
 291 section depends on one. The class is now inhabited — `Examples.lean` §11
 292 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
 293 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
 294 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
 295 changed by the symmetry, so an energy-minimising field and its image have
 296 equal energy and differ. That is symmetry breaking exhibited rather than
 297 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 298 no one-parameter family to differentiate along in the first place. Simultane-
 299 ously, discrete permutation symmetries dictate the statistical mechanics of
 300 any physical system, defining a strictly finite volume of phase space.¹

301 Because phase space is finite, any localized physical system possesses a

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
 amples, the underlying Lean formalization does not hardcode specific physics. It defines
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-
 ics). What the formalization supplies is the vocabulary in which symmetries, conserved
 quantities and phase-space constraints can be *stated*; it does not derive them. All three of
 the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invari-
 ance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses
 of inhabitability, not of physics: no theorem in the development consumes a symmetry
 group or a metric, so this section still contributes vocabulary rather than content.

302 mathematical limit on the amount of information (distinguishable states) it
303 can embody at any given moment.

304 4 Derivation 1: Symmetry Breaking and the 305 Inevitability of Boundaries

306 A perfectly symmetric universe is the uniform vacuum; no distinct entities
307 exist. For any dynamic process to occur, the continuous symmetries of the
308 background field must undergo spontaneous symmetry breaking.

309 When a field drops to a lower, asymmetric energy state, topology dic-
310 tates the natural creation of defects (such as domain walls or monopoles).
311 This constraint necessitates the creation of a physical boundary—a localized
312 topological distinction between an “inside” and an “outside.” While such
313 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
314 entity, it is not sufficient for agency. The boundary is relational, defined by
315 the twisted geometry of the field surrounding it, and must be coupled with
316 non-equilibrium thermodynamics to exhibit active behaviors.

317 That first sentence was, until recently, the one claim of this section that
318 the formalization did not make. What Lean established was the energy half
319 — an energy-minimizing field sits in the vacuum manifold almost everywhere
320 (`pointwise_vacuum_of_global_min`) — together with the observation that
321 a boundary defect *obstructs* extension over a contractible interior. Neither
322 produces a defect, so “topology dictates the natural creation” was asserted

323 rather than proved. It is now the theorem `exists_notMem_of_connected-`
 324 `ComponentIn`: if the substrate is connected and a continuous field takes val-
 325 ues in two *different connected components* of the vacuum manifold M , then
 326 somewhere the field is not in M at all. That excluded region is the wall.
 327 This is the π_0 case of Kibble’s classification of topological defects by the ho-
 328 motopy of the vacuum manifold (Kibble, 1976), and the proof is elementary
 329 in exactly the way it is elementary in the physics literature: the continuous
 330 image of a connected space is connected, a connected subset of M lies in-
 331 side one component of M , and the field’s range would be such a subset if it
 332 never left M . No hypothesis mentions energy, a potential, or a symmetry;
 333 connectedness of the substrate and disconnectedness of the vacuum are the
 334 whole of it.

335 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 336 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 337 classification and Mathlib’s homotopy-group library would support at least
 338 the next one; they are available and not taken, because the argument this
 339 paper runs uses domain walls and nothing else. The witness is the double
 340 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be
 341 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 342 minima, and its disconnection is proved in the form the theorem consumes:
 343 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 344 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
 345 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field

346 on *any* connected substrate that reaches both minima, with no formula for
 347 the field anywhere in the statement. That the class is non-empty is checked
 348 separately, because a theorem quantified over all fields is worthless if none
 349 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 350 forced to have sits at the origin (`kink.zero.notMem`).

351 **5 Derivation 2: The Thermodynamic Con-** 352 **straint**

353 This newly formed topological boundary is subjected to gauge perturbations
 354 from the “outside” environment. Because the internal phase space is finite
 355 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
 356 and must overwrite old internal states.

357 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
 358 formation irreversibly decreases internal entropy, mandating that the sys-
 359 tem exhaust heat into the environment. Philosophers of physics (such as
 360 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
 361 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
 362 (a many-to-one mapping in abstract computational states) with *thermody-*
 363 *namic* erasure (a many-to-one mapping in physical phase space). They argue
 364 that logical erasure could conceptually be implemented by an injective phys-
 365 ical process without dissipating heat.

366 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)

367 addresses this critique by defining erasure as a non-injective mapping on the
 368 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
 369 tion of the bipartite universe (System + Bath) as strictly injective, repre-
 370 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
 371 to abstract logical erasure, but is a mandatory mathematical consequence of
 372 any local physical state space contraction under globally reversible dynamics.
 373 If a cognitive system’s internal topology forces a contraction of its physical
 374 state space to accommodate structural resonance, heat dissipation becomes a
 375 thermodynamic requirement. A stable boundary thus constitutes a dissipa-
 376 tive structure (Prigogine & Stengers, 1984), operating as a non-equilibrium
 377 steady state that consumes free energy to pay the thermodynamic cost of
 378 erasure. Continuous thermal fluctuations play a constructive role in estab-
 379 lishing boundaries between states while contributing to the system’s flexible,
 380 adaptive response.

381 **6 Derivation 3: Prediction as the Thermody-** 382 **namic Cost of Memory**

383 A dissipative structure is driven by its environment. Write X_t for the state
 384 of the structure and S_t for the environmental signal driving it, and let the
 385 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
 386 does not act. Two quantities are then defined at each step: the *memory*
 387 $I(X_t; S_t)$, the mutual information between the structure’s present state and

388 the signal's present value, and the *predictive information* $I(X_t; S_{t+1})$, which
389 is what that same state says about the signal's next value.

390 **What this section used to claim, and why it is withdrawn.** Earlier
391 versions of this derivation argued as follows. Model the environment as a
392 stochastic driving force $P(ext)$ and the structure's internal transitions as
393 $Q(int)$; entropy production σ is then bounded below by $D_{KL}(P||Q)/\Delta t$;
394 the principle of least action drives σ downward; therefore $D_{KL}(P||Q) \rightarrow$
395 0, and the internal transition probabilities come to mirror the statistical
396 symmetries of the environment. The final step does not follow. Rearranged,
397 the bound reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so descending on σ lowers a *bound* on
398 the divergence; but a driven system does not relax to $\sigma = 0$. It relaxes to
399 a non-equilibrium steady state at strictly positive entropy production, and
400 the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number. Nothing
401 forces the divergence to zero. The inequality itself is true, is formalized as an
402 instance obligation, and does real work — combined with Gibbs' inequality
403 it gives $\sigma \geq 0$ — but the conclusion drawn from it was not licensed, and
404 it was that conclusion which supplied this framework's entire connection to
405 predictive processing.

406 **The replacement is a published theorem.** Still, Sivak, Bell and Crooks
407 (Still et al., 2012) prove that for a system driven by an environmental signal

408 the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

409 The bracket is the *nonpredictive* information: what the structure retains
410 about the signal's present that says nothing about the signal's future. Dis-
411 sipation is bounded below by it. Memory that fails to predict is not merely
412 useless to the structure; it is thermodynamic waste, and the waste is quan-
413 tified in energy.

414 **Three consequences, and they are theorems rather than readings.**
415 The bound is carried in Lean as a class field, `PredictiveDissipation`.
416 `still_bound (Phase3.PredictiveThermodynamics)`, for the same reason ev-
417 ery other physical postulate here is: its `axiom` form, quantified over all joint
418 laws with the dissipated work fixed, is refutable, and **Examples** §18.4 refutes
419 it. What is *derived* is everything downstream.

420 First, *the bracket is non-negative*, and this is proved rather than assumed.
421 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
422 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
423 $I(X_t; S_t)$: a system cannot know more about the signal's future than about its
424 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback–
425 Leibler divergence but no mutual information, so the latter is defined here
426 as the divergence of the joint law from the product of its marginals, which is
427 the standard definition and makes the library's data processing lemma apply

428 directly.

429 Second, *the second law follows* instead of being assumed. Dissipation is
430 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$
431 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays for
432 the older bound, with one difference that matters: Gibbs' inequality com-
433 pares a system to an arbitrary reference distribution, whereas the data pro-
434 cessing inequality compares the system to its own future, which is a quantity
435 selection can act on.

436 Third, and this is the point of the replacement, *the inference now runs*
437 *in the valid direction*. The quantity of interest sits on the small side of
438 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
439 Any bound on dissipation is immediately a bound on wasted memory, so a
440 structure held to an energy budget is thereby held to a prediction standard:
441 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case
442 $W_{\text{diss}} = 0$, the theorem `predictive_eq_of_no_dissipation` gives that every
443 retained bit is predictive. This is a weaker statement than the one it replaces,
444 and it is the one that is true.

445 **The hypothesis is load-bearing, and we exhibit what breaks with-**
446 **out it.** The data processing step requires that the structure not write to
447 its environment. In the formalization this is not an assumption stated beside
448 the theorem but the shape of the evolution itself: the signal's step is applied
449 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the

450 system coordinate and through κ on the signal coordinate, so no channel
 451 from X_t to S_{t+1} exists. **Examples §18.3** shows the restriction is not decora-
 452 tion: a Markov kernel on the *pair* which copies the system’s state into the
 453 signal carries an independent two-bit law to a perfectly correlated one, rais-
 454 ing mutual information from zero to a positive value. An agent that steers
 455 its environment can indeed know more about that environment’s future than
 456 about its present, and Still’s bound is not about such an agent.

457 **Both regimes are exhibited, not just described.** **Examples §18** builds
 458 two systems over one correlated two-bit joint law, differing only in the signal’s
 459 dynamics. When the signal is frozen, every remembered bit is a predictive
 460 bit, the nonpredictive information is zero, and the instance is permitted to
 461 dissipate nothing — with the memory itself proved strictly positive, so the
 462 conclusion is about a system that does remember something. When the signal
 463 is redrawn from its stationary law at each step, the predictive information is
 464 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
 465 to dissipate $k_B T$ times its whole mutual information. Establishing that the
 466 mutual information of the witness is neither zero nor infinite is where the
 467 work of that section lies.

468 **An equality, where the old argument had only a bound.** Kawai,
 469 Parrondo and Van den Broeck (Kawai et al., 2007) show that the average
 470 dissipated work is exactly $k_B T$ times the relative entropy between the for-
 471 ward and time-reversed phase-space densities. Carried here as a predicate on

472 an instance rather than as a further postulate, it replaces the abstract dis-
473 sipation field with a quantity an experiment can estimate — the statistical
474 distinguishability of a recording of the system from the same recording run
475 backwards — and yields the statement that the irreversibility of the dynamics
476 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
477 `le_arrow`).

478 **What this does not establish.** It does not deliver the Free Energy Prin-
479 ciple. Still’s bound says that prediction is thermodynamically favoured; it
480 says nothing about hierarchical generative models, variational free energy,
481 or Bayesian inference, and we do not claim to have derived any of them.
482 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
483 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-
484 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
485 proved here, which is a weaker relation than derivation and the only one
486 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
487 Maximal predictive information is compatible with the structure storing a
488 lossy, compressed or otherwise unrecognisable function of the signal; a state
489 that predicts is not thereby a state that means, and earlier versions of this
490 section overstated the case. What survives of that argument is narrower and
491 still worth saying: whatever internal states a physical structure holds, those
492 that do not help predict its environment are paid for in heat, so the pressure
493 toward states indexed to the environment is energetic rather than semantic.

494 We continue to call the phenomenon *structural resonance*, and there is no
495 teleological drive to “survive” in it: predicting the environment is simply the
496 cheaper of the two ways to hold a memory.

497 **7 Macroscopic Scaling: A Theorem and a** 498 **Commitment**

499 This is the chain’s fourth step, and it is deliberately not numbered as a deriva-
500 tion. The change is not cosmetic. Earlier drafts ran two steps of different
501 kinds together in a single paragraph: a coarse-graining argument, which is
502 a piece of mathematics, and the identification of the coarse-grained medium
503 with the brain’s endogenous electromagnetic field, which is an empirical claim
504 about cortex. Presenting the second as though it inherited the licence of the
505 first made it look like a deduction, and it is not one. Separated, the empirical
506 claim becomes the framework’s central commitment and the place where it
507 is most easily wrong — which is where a claim of this kind belongs.

508 **The formal half: coarse-graining is a theorem, and a** 509 **narrow one**

510 A network of discretely coupled units has a coupling energy that is a sym-
511 metric double sum over pairs. `Phase2.MeshConvergence.lean` proves that
512 this sum converges to a continuous integral: along a sequence of regular

513 triangulations of a region S whose edge regions shrink to points, the dis-
 514 crete energy $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$ converges to $\int_S f d\mu$ (`mesh_refinement_`
 515 `convergence`), witnessed on the uniform partition of $[0, 1)$ in Examples §6.
 516 The identification that makes this a coarse-graining rather than an arbitrary
 517 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,
 518 the double sum is exactly the midpoint rule over unordered edges, so the
 519 discrete energy is a quadrature of the continuous one rather than a different
 520 quantity that happens to converge to it.

521 What this establishes is worth stating exactly, because it is less than the
 522 phrase “coarse-graining, governed by the Renormalization Group” suggests.
 523 It is a statement about an *energy functional*, not about a dynamics: noth-
 524 ing here shows that trajectories of the discrete Kuramoto system converge
 525 to solutions of a field equation, and the earlier appeal to the Renormaliza-
 526 tion Group was decorative — no renormalization group flow is constructed
 527 anywhere in this development. The limit object is a scalar density on the sub-
 528 strate, in the intended reading the magnitude of the stress-energy tensor, and
 529 it is not an electromagnetic field. The continuous object the later sections
 530 use, `ContinuousNeuralField`, is *posited* as a structure with a coupling kernel
 531 K ; no theorem produces it from the discrete system. The gap between the
 532 finite Kuramoto model and its continuum limit is the propagation-of-chaos
 533 problem, which we have not addressed and record as open.

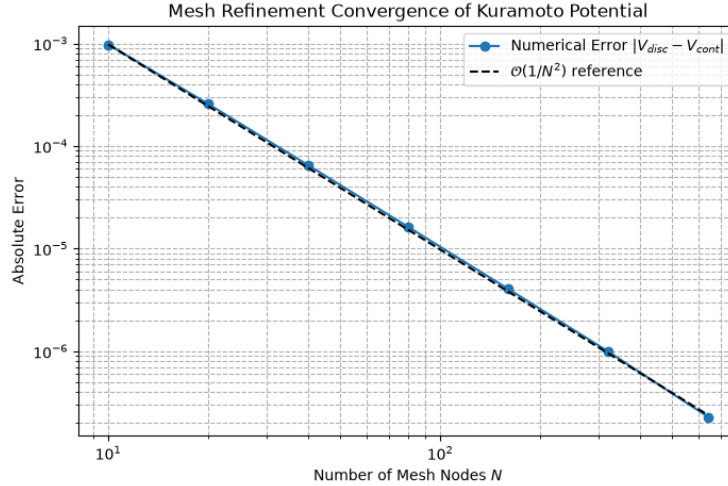


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones.

534 **The empirical half: which physical field realizes the ker-**
 535 **nel**

536 The formal half says that a large network of symmetrically coupled units
 537 admits a continuum description with a symmetric kernel. It says nothing
 538 about which physical field in cortex that kernel describes. Our answer is an
 539 empirical commitment, and we state it as one:

540 *In the mammalian cortex, the field realizing the coarse-grained*

541 *coupling kernel is the endogenous electromagnetic field.*

542 This is a claim about biology, not a consequence of anything above it.
543 It is what makes the framework falsifiable and it is what we expect to be
544 attacked.

545 **What supports it.** Neurons do not only signal through synapses: trans-
546 membrane currents generate extracellular fields that shift the membrane po-
547 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-
548 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike
549 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic
550 head models put mesoscopic ephaptic effects within the range where they
551 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic
552 treatments show field coupling is sufficient for coincidence detection in the
553 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have
554 been argued to carry information that survives representational drift in the
555 underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha &
556 Lima, 2024). The field is spatially continuous and superposes contributions
557 from large populations, which is the property the framework needs: it is a
558 medium in which spatially separated activity is combined without a further
559 mechanism for combining it. This is the integrative role the CEMI field
560 theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

561 **What the evidence does not support, and a claim we withdraw.**
562 Earlier drafts asserted that *the continuous macroscopic EM field is the pri-*
563 *mary dissipative structure interacting with the universe.* The word “primary”
564 is not supported by any of the work cited above and we withdraw it. What
565 the experiments show is *modulation*: Anastassiou and Koch report that en-
566 dogenous fields of 1–5 mV/mm shift spike timing by one to three millisec-
567 onds (Anastassiou & Koch, 2015), an effect that is real, reproducible, and
568 subthreshold. It biases when a cell fires; it does not decide whether it fires.
569 The corrected claim is that the endogenous field is *modulatory in its effect on*
570 *individual neurons and integrative in its spatial reach* — weak per synapse,
571 but the only variable in the system that is defined everywhere at once. The
572 framework needs the second property, not the first, and conflating them was
573 the error.

574 **Its critics.** The objection with the longest history is that the fields are
575 too weak to matter, and it has been pressed from within the field-theoretic
576 camp: Pockett, who has herself defended an electromagnetic theory of con-
577 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the
578 measurement problem of relating gross field structure to reported experience,
579 the absence of a constant relationship between the two, and the question of
580 whether the field does causal work or merely reflects it. The quantitative case
581 for scepticism has strengthened since. Vöröslakos and colleagues measured
582 intracranial fields during transcranial stimulation in rats and human cadav-

ers and found that the great majority of applied current is shunted through the scalp, so that conventional protocols deliver fields at or below the 1 V/m range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields rather than endogenous ones, but it constrains how much causal work a field of that magnitude can be doing, and it is the reason we do not claim the field drives spiking.

What would falsify it. We state the failure conditions rather than leaving them to a referee.

1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted — by a conductive medium that nulls the extracellular potential without altering transmembrane currents — and neither behaviour nor report changes, the identification fails. This is the decisive experiment and it is, in principle, available.

2. *A magnitude gap.* If the spatial integration the framework requires demands field effects an order of magnitude above the 1–5 mV/mm measured in cortex, the commitment fails on arithmetic. The relevant number is not whether ephaptic coupling exists but whether its coupling strength suffices to place the system above the critical coupling K_c of Derivation 7.

3. *A timescale mismatch in the wrong direction.* The prediction of Section 11 turns on the field’s dynamics being fast relative to the astrocytic

604 and clearance variables. If recovery from sleep inertia tracks an expo-
605 nential set by the electrical timescale rather than the delayed sigmoid
606 the slow variable predicts, the framework loses the one prediction that
607 is its own.

608 4. *Dissociation of field coherence from reported unity.* If large-scale field
609 coherence is preserved across states in which experiential unity is lost
610 — or absent in states where it is retained — the identification of the
611 global section with unified experience has no physical carrier.

612 A framework that survives all four is not thereby true. It is merely still
613 standing, which is more than most theories of consciousness arrange to be
614 able to lose.

615 **Geometric frustration, and the regime our theorems ac-** 616 **tually cover**

617 Earlier drafts grounded the system’s memory capacity in *geometric frustra-*
618 *tion*: a highly coupled network cannot bring all its nodes to their lowest en-
619 ergy simultaneously, producing a jagged landscape of near-degenerate states
620 with the capacity to store many patterns. That argument is standard and it
621 is not what this development formalizes. Confronting the discrepancy is the
622 honest course, so we do it here rather than in a footnote.

623 Frustration requires competing couplings — in the Kuramoto setting, a
624 sign structure that no phase assignment can satisfy on every edge at once.

625 The formal setting admits this: `KuramotoSystem` carries $A : V \rightarrow V \rightarrow$
 626 $\mathbb{R}$, symmetric and otherwise unconstrained, so signed couplings are repre-
 627 sentable. But every theorem in this development that says anything about
 628 *where* the dynamics ends requires strictly positive coupling. `phase_locked_`
 629 `minimizes_potential` assumes $A_{ij} > 0$; `potential_min_iff_phase_locked`
 630 inherits it; the class field `ThermodynamicCover.A_pos` demands it of every
 631 instance, and hence of every conclusion of Derivation 5; and the convergence
 632 theorem of `Phase4.RotatingFrame` §7 requires the strictly stronger uniform
 633 bound $0 < a \leq A_{ij}$. A network with all-positive couplings is an unfrustrated
 634 ferromagnet whose potential has one minimum up to a global phase shift. It
 635 is the opposite of a jagged landscape, and it has no capacity to store multiple
 636 patterns at all. **The system in which memory is grounded and the**
 637 **system our theorems describe are not the same system.**

638 The division is not arbitrary, and it is worth recording where it falls. What
 639 survives with no assumption on the sign of the coupling is the dynamical half:
 640 well-posedness on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`, `_unique`), co-
 641 variance under passage to a rotating frame, monotone descent of the potential
 642 (`dynamic_potential_antitone`), finiteness of the dissipation integral, and —
 643 through Barbalat’s lemma — that the motion stops along every trajectory
 644 (`velocity_sq_tendsto_zero`). *That* the motion stops is frustration-agnostic.
 645 *Where* it stops is not, and every claim this framework makes about the des-
 646 tination lives on the positive side of that line. On the continuum side the
 647 accounting differs slightly: `phase_locked_minimizes_entropy_of_symm` as-

648 sumes only that the kernel is symmetric and so tolerates signed kernels, but
649 it is conditional on a dynamically phase-locked field existing — which is
650 precisely what frustration may prevent.

651 We therefore scope the claim rather than repeat it. The memory capacity
652 of a frustrated cortical network is not supported by any theorem here, and we
653 do not assert it as part of the chain. What the chain uses is the unfrustrated
654 regime, in which synchronization is reached and the global section exists.
655 Whether a frustrated network can hold many near-degenerate patterns *and*
656 still synchronize enough of itself to admit a global section is an open question
657 about mixed-sign oscillator networks, and it is the most substantive gap we
658 know of in this framework. Recording it as such costs a claim we would have
659 liked to make and is preferable to holding two incompatible pictures at once.

660 **The supporting physiology**

661 The continuous medium is physically supported by astrocytic syncytia—non-
662 neuronal networks joined by gap junctions that provide macroscopic cytoplas-
663 mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).
664 Managing the thermodynamic cost of this state requires sophisticated micro-
665 bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-
666 tochondrial Information Processing Systems (MIPS), transducing biological
667 and environmental information to sustain the coherence of the macroscopic
668 field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled with mito-
669 chondrial micro-mechanics, generate and sustain the field. Because the exter-

670 nal environment is continuous, achieving maximal thermodynamic structural
671 resonance is optimally realized within this continuous macroscopic medium
672 (van Bree et al., 2025).

673 8 Derivation 5: Global Sections and the Unity 674 of Consciousness

675 The system is now characterized by a continuous macroscopic electrodynamic
676 field generated by millions of coupled micro-components, forced to resonate
677 with the environment. Following Section 7, the coupling regime assumed
678 from here on is the unfrustrated one — all couplings positive — which is
679 what every theorem below requires and what the frustrated network of the
680 memory argument is not.

681 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
682 tors guarantees that, past a critical coupling threshold, these distributed
683 local fields undergo a phase transition into unified synchronization (Bauer et
684 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
685 phase-locked state minimizes the Lyapunov potential of the system, in the
686 precise sense discussed in Section 2.1: the functional minimized is the cou-
687 pling potential read in the frame rotating with the natural frequency, which
688 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
689 system of identical oscillators.

690 This phase transition mathematically glues local synchronized states into

691 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
 692 The EM field acts as a single, indivisible topological space, providing the
 693 *Unity of Consciousness*—a necessary condition for integrated experience that
 694 solves the binding problem. While Global Workspace Theory (Baars, 1997;
 695 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 696 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 697 our derivation provides a continuous physical mechanism. The contrast is
 698 worth drawing carefully, because both theories are addressing the same ex-
 699 planandum from different directions.

700 We state the scope of the Lean result precisely, because it is narrower
 701 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 702 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 703 `phase_locked` characterizes those minimizers exactly: they are the phase-
 704 locked configurations, in both directions. Every instance is therefore already
 705 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
 706 condition on its local data — that patches at a common phase agree where
 707 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 708 sections compatible. `probability_glue_unique`, which is the sheaf property
 709 of the probability presheaf and contains no physics at all, then glues that
 710 family into exactly one global section.

711 An earlier version of this structure proved less than the prose claimed,
 712 and we record the correction because it changed what the theorem says
 713 rather than how it is stated. `LocalSectionSynchronization` used to carry

714 a *global* section as a class field, together with a second field declaring every
 715 local section to be a restriction of it. The global object therefore existed
 716 before the sheaf condition was ever consulted, and `global_section_from_`
 717 `thermodynamics` could do no more than rule out competitors to an object
 718 the instance had supplied. Both fields are gone. Every field of the class
 719 now mentions a single patch or a single overlap; the global section is con-
 720 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
 721 survives as a *theorem* about that construction (`ThermodynamicCover.sync_`
 722 `to_section_eq`), so the local sections are restrictions of one common measure
 723 because the sheaf condition says so, not because an instance was required to
 724 name one.

725 `Examples.lean` §14 is the witness that the difference is not notational.
 726 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 727 the three sites, agreeing only at the site they share, where both put mass 1
 728 — which is what discharges the overlap condition, by computing two num-
 729 bers rather than by construction. The section the theorem returns has pro-
 730 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
 731 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
 732 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 733 tion that no single patch had. `Examples.lean` §13, built the older way
 734 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 735 is retained alongside it for the negative result it carries: its phase field is not
 736 the constant function — the second patch a full turn ahead of the first — and

by `ThermodynamicCover.phase_locked` that is as far from constant as any instance can get, since no instance can carry phases differing by anything a measure is allowed to see.

Two hypotheses carry physical content, and until recently each was an obligation on every instance. One still is: that synchronized patches agree on their overlaps is a modelling assumption, not a consequence of any dynamics we formalize. The other, `thermodynamic_equilibrium`, asks that the cover already sit at the potential minimum, and no theorem says an *arbitrary* system’s dynamics takes it there. “A thermodynamic phase transition into unity” is therefore still doing causal work that no general Lean result supports, though it is no longer entirely without one: `Examples.lean` §15 exhibits a two-oscillator trajectory that starts a quarter turn out of phase and converges to the global minimum of the dynamic potential, with the order parameter tending to 1. That trajectory is not an isolated construction: `kuramoto_cauchy_problem` proves the Kuramoto initial value problem well posed on all of $\mathbb{R}$, so through every initial state there passes exactly one trajectory, and the exhibited one is the only solution through its own. Nor is its settling down peculiar to it: along every trajectory of a zero-frequency system the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite and equal to the potential’s total drop (`dynamic_potential_tendsto`, `dissipation_integral_tendsto`). The motion also stops: a finite integral does not by itself force its integrand to zero, but the dissipation rate is Lipschitz in time, so Barbalat’s lemma applies, and every trajectory satis-

760 $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it
 761 is proved here as `tendsto_zero_of_lipschitz_of_integral_le`). That last
 762 step — locating where the motion stops — is now taken, though condition-
 763 ally, and by a different route than we had recorded. We had marked it as
 764 blocked on the LaSalle invariance principle, which Mathlib does not have,
 765 and on the absence of a compact invariant set in the state space $V \rightarrow \mathbb{R}$.
 766 That was the wrong diagnosis. LaSalle is the tool for a system with a de-
 767 creasing functional and nothing else; this one has more. The flow is a gra-
 768 dient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather
 769 than an inequality; the minimisers of V are characterised exactly; and on
 770 the region where every pair of phases lies within a quarter turn a Polyak-
 771 Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is the poten-
 772 tial's excess above its minimum and $a > 0$ is a lower bound for the coupling
 773 (`lojasiewicz_estimate`). Its proof is two extremal indices and the iden-
 774 tity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase is
 775 pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 776 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 777 as well, because the invariant region is produced by the Lyapunov function
 778 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 779 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 780 continuity on the connected half-line then prevents any pair from chang-
 781 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 782 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each

783 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 784 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 785 `minimum`: if the phases start within a quarter turn of each other and the initial
 786 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 787 which is a *global* minimiser of the dynamic potential, with order parameter
 788 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 789 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
 790 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 791 tion: the trajectory comes from the existence theorem and its limit from the
 792 convergence theorem, so neither is written down.

793 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 794 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 795 cover whose phase field is the limit of such a trajectory and returns an in-
 796 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 797 *vergence*; the cover's coupling matrix is the coupling matrix of the dynamics
 798 rather than a separately chosen one, so the system whose equilibrium the
 799 class asserts and the system whose trajectory is run are the same system.
 800 `Examples.lean` §17.1 is the resulting witness: three patches over the three-
 801 site substrate, each carrying its own mass profile on its own open set, above a
 802 phase field that no line of the construction writes down. The trajectory that
 803 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 804 `start_not_locked`), so on this cover the configuration Derivation 5 requires
 805 is arrived at rather than posited. The state the three patches glue to has

806 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
807 `ne_glued`), so one cover now carries both the emergence reading of the gluing
808 and the derived reading of the equilibrium. What is still assumed on it is the
809 overlap agreement, and nothing else.

810 The two hypotheses cannot be dropped, and this is the honest scope of
811 the result. Splay and twisted configurations are equilibria of the same flow,
812 so “every trajectory reaches the unified state” is false as a general statement,
813 and the initial-data condition is the standard arc condition under which the
814 true statement holds. What is claimed is therefore conditional — unity is
815 reached from configurations already sufficiently coherent — and not that an
816 arbitrary initial state is carried into unity. What Lean establishes about the
817 gluing itself is the other half: *given* locally synchronized patches, a global
818 state exists, is unique, and is produced by the gluing rather than presupposed
819 by it.

820 Integrated Information Theory begins from phenomenological axioms and
821 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
822 with the exclusion postulate selecting a single maximum- Φ complex as the
823 locus of experience. Our sheaf-theoretic global section plays a structurally
824 analogous role — it too picks out a distinguished unified object — but it
825 is arrived at from the opposite end: not postulated from the character of
826 experience, but forced by a thermodynamic phase transition in a physical
827 field. This buys two things IIT does not have. First, the unity condition
828 is checkable from field dynamics rather than requiring the computation of

829 Φ , which is intractable for systems of realistic size and whose definition has
830 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
831 out rather than being postulated: gluing either succeeds on a given cover or
832 it does not, so there is no separate principle needed to rule out overlapping
833 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
834 that Φ can be high in systems with no plausible claim to experience, such
835 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
836 to the mirror objection, since a physical system could in principle satisfy
837 the gluing condition without being cognitively interesting. We take this
838 seriously rather than dismissing it: our reply is that gluing is necessary but
839 not sufficient, and that the topological boundary of Derivation 1 and the
840 reflexive structure of Derivation 6 are additional required conditions. IIT, by
841 contrast, treats Φ as both necessary and sufficient.

842 Global Workspace Theory faces a complementary difficulty. It specifies
843 a functional architecture — a limited-capacity workspace broadcasting to
844 specialized modules — without specifying what physical property of broad-
845 casting makes the broadcast content conscious, leaving the theory open to the
846 charge that it explains access consciousness while leaving phenomenal con-
847 sciousness untouched (Block, 1995). Our framework can be read as supplying
848 the missing physical substrate for the workspace: the continuous macroscopic
849 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
850 old gives a physical condition for ignition rather than a functional stipulation
851 of it. Where GWT is agnostic about implementation, we make a falsifiable

852 commitment — unity requires a continuous physical medium, which predicts
 853 that a functionally equivalent discrete broadcast architecture would not be
 854 conscious. We regard this as a genuine cost of the position, not a free ad-
 855 vantage: it is the point at which our framework departs most sharply from
 856 computational functionalism and where it is most readily falsified. Contrast
 857 this derivation with panpsychism: consciousness is not assumed as a funda-
 858 mental property of matter. It requires the initial existence of a topological
 859 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 860 dynamic phase transition into unity that occurs when integrated spatial cou-
 861 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 862 the foundational topological boundary and the subsequent mathematically
 863 defined phase transition, the system remains a fragmented, non-conscious
 864 collection of micro-states. A unified field of perception, however, is not yet
 865 a “Self.”

866 **9 Derivation 6: Reflexive Topology and the** 867 **Emergence of the Self**

868 While a global section provides Unity, a “Self” requires reflexivity. The
 869 unified EM field is a dissipative structure that must resonate with external
 870 perturbations while managing its own internal phase transitions. Time evo-
 871 lution naturally drives the field to minimize thermodynamic friction with its
 872 own internal state-changes.

873 This mandates *auto-resonance*: the field must predict itself. The “Self”
 874 emerges as a higher-order topological feature when the internal geometry
 875 folds to include a low-dimensional avatar of its own computational and causal
 876 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)
 877 (Levin, 2019). The system is unified, predictive, and explicitly maps its
 878 own distinct existence from the environment. This architecture provides the
 879 physical foundation for a self-aware consciousness, bounded by the continuous
 880 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).
 881 We are explicit about what carries the weight here. `reflexive_topology_implies_self`
 882 is the Banach fixed-point theorem with the physical commitments in its hy-
 883 potheses: the space of global sections is assumed nonempty, metric and com-
 884 plete, and the predictive map is assumed to contract by $1/2$. `Examples.lean` §10
 885 discharges all four on the three-site substrate of Derivation 5, and the metric
 886 is no longer a placeholder. The section first pierces the sheafification: on
 887 a discrete substrate every point has a smallest open neighbourhood, so the
 888 mass a germ carries at a point is well defined, and `massEquivOn` proves that
 889 over *any* open the sections of the probability sheaf *are* the finite measures
 890 on that open, read as densities against counting measure — the global case
 891 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 892 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 893 correspondence had to be constructed. The metric on sections is then the
 894 uniform distance between the densities of the measures they glue to — equiv-
 895 alent to their total variation on a finite substrate, though not equal to it —

896 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 897 of densities, not by the degenerate route of showing Cauchy sequences are
 898 eventually constant. The self-prediction map is the relaxation “average the
 899 current field with the baseline,” and we prove it contracts distances by *ex-*
 900 *actly* one half, that it is not constant, and that its fixed point is unique. An
 901 earlier version of this witness used the 0/1 metric, under which every con-
 902 traction is constant; that theorem is kept in the source as the record of why
 903 the earlier witness was empty. A separate gap, closed at the same time, con-
 904 cerned the avatar rather than the fixed point. `ReflexiveBoundary` carries a
 905 field `auto_resonance` sending the global state to a section over the avatar re-
 906 gion, and nothing constrained it: a constant function was a legal avatar, and
 907 `reflexive_topology_implies_self` consumes only the predictive model, so
 908 no theorem in the development mentioned the field the avatar is supposed to
 909 be reading. The constraint is now stated as a predicate rather than added
 910 to the structure — `IsRestrictionResonance`, that the avatar’s state is the
 911 presheaf restriction of the global state — and two theorems say why it is
 912 the right one. In the positive direction, if a family of resonant avatars *cov-*
 913 *ers* the substrate then the global state is determined by their local readings
 914 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 915 recovered from them, by the same sheaf condition that Derivation 5 uses.
 916 In the negative direction, replacing any avatar by a constant leaves a legal
 917 `ReflexiveBoundary` with the same region, the same predictive model and
 918 the same Self (`self_of_constResonance`), so the fixed-point theorem cannot

919 see the difference and the predicate is doing the work rather than restating
 920 it. Both halves are witnessed on the three-site substrate, where the avatar
 921 region provably separates two states of the field, so the condition is a con-
 922 straint there and not a formality. What is still not established is that any field
 923 dynamics produces this particular map or performs the restriction, or that
 924 the contraction constant $1/2$ is read off an entropy production rate rather
 925 than built into the map's definition. We also note a negative result found
 926 in the course of this work: the Lévy–Prokhorov metric, which the natural
 927 weak-convergence reading would suggest, does *not* support this argument on
 928 a discrete substrate, because the ε -thickening of a set is the set itself for $\varepsilon < 1$
 929 and the relaxation map is consequently not a Lévy–Prokhorov contraction.

930 **10 Derivation 7: Continuous Stochastic Ther-** 931 **modynamics and Field Theory**

932 To move beyond finite graph analogies, we model the macroscopic electro-
 933 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
 934 continuous spatial phenomenon, not a discrete network. We extend the
 935 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
 936 (`Phase8ContinuousField.lean`).

937 Subject to thermodynamic noise D , the system undergoes non-equilibrium
 938 stochastic thermodynamics. Rather than a static Landauer bound, the field

939 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

940 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
 941 dergo gradient descent on this entropy production functional.

942 That gradient descent is a theorem rather than a modelling assertion,
 943 for finite substrates. Reading σ as a functional of the coupling kernel at
 944 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
 945 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

946 and `sigmaOfKernel_eq_entropy_production_rate` identifies the functional
 947 being differentiated with `entropy_production_rate` itself whenever the sub-
 948 strate carries counting measure. `structural_resonance_decreases_entropy_production`
 949 then concludes that σ is non-increasing along any coupling trajectory relax-
 950 ing down its own gradient at a positive rate c . Earlier drafts proved this
 951 only for an abstract functional S on an abstract inner-product space with
 952 the gradient supplied as data; the identification of that S with σ was infor-
 953 mal, and Table S1 flagged the row accordingly. Two limitations stay explicit,
 954 and the first is narrower than we previously reported. The derivation is for
 955 *finite* substrates. We had recorded the obstacle to the continuum case as

956 differentiation under the integral sign with respect to the kernel; that was
 957 a misdiagnosis. σ is not a general functional that happens to be written as
 958 an integral — it is a quadratic functional of an *affine* image of the kernel,
 959 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
 960 constant times the squared norm of the drift. A squared norm is Fréchet
 961 differentiable on any real inner-product space and an affine map is its own
 962 derivative, so the composite differentiates by the chain rule with no limiting
 963 argument. `hasFDerivAt_quadratic_of_affine` states that in full generality,
 964 and `hasFDerivAt_sigmaOfKernel_of_operator` re-derives the finite gradient
 965 from it, showing that the closed form displayed above is a convenience rather
 966 than the content. What the continuum case still needs is therefore one object:
 967 the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded operator
 968 $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral ker-
 969 nel is bounded by 1, so on a finite measure space the operator norm is at most
 970 $\mu(M)^{1/2}$. We have not built it, because it requires constructing a continu-
 971 ous linear map between L^p spaces and we have not done that bookkeeping;
 972 we record the corrected scope because “construct one bounded operator”
 973 and “differentiate under the integral sign” are different sizes of task. The
 974 second limitation is unchanged: the phase field is held fixed throughout, so
 975 this is plasticity of the coupling at frozen phases, not joint (θ, K) dynamics.
 976 **Examples.lean** §5 exercises the whole chain on a two-site substrate whose
 977 coupling relaxes as e^{-t} and whose entropy production decays as e^{-2t} , con-
 978 firming that the gradient is not identically zero and that the descent is strict

979 rather than vacuously monotone.

980 A second result about σ concerns where the descent is headed, and it has
981 been strengthened since the last draft in a way that also corrects an estimate
982 we had recorded. That the phase-locked field minimizes σ was proved only
983 against competitors sharing its spatial mean drift — a restriction that left
984 the theorem close to Jensen’s inequality and no more — and we had judged
985 that lifting it would require modelling how the mean drift varies with the
986 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
987 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
988 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
989 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
990 its own negative and the total coupling contribution vanishes. The total drift
991 $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
992 `symm`): coupling moves drift between sites but neither creates nor destroys it.
993 Evaluating that conserved quantity at the phase-locked configuration, where
994 the drift is constant, pins its value, and the mean-matching hypothesis follows
995 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
996 restricted statement. Reciprocity is the physically intended condition — an
997 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
998 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
999 not a formality, and we prove that rather than assert it. `Examples.lean` §16
1000 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
1001 feel a , whose phase-locked field is strictly beaten: displacing the second phase

1002 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
 1003 ond, halving σ . So “phase-locking minimizes entropy production” is false for
 1004 general kernels and true for reciprocal ones, and the hypothesis cannot be
 1005 removed by working harder.

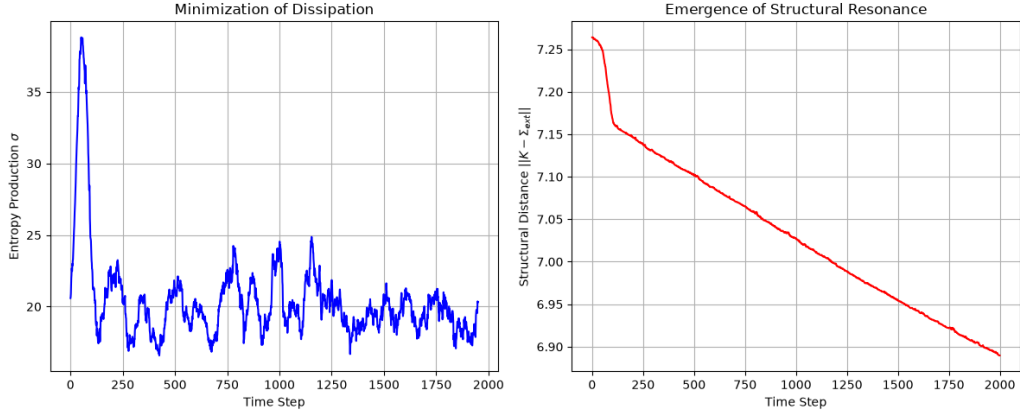


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason Section 10 withdraws the old inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence.

1006 We define the critical coupling threshold $K_c = 2D$. The phase transition
 1007 to the unified “Global Section” occurs when the mean-field spatial coupling
 1008 exceeds this thermodynamic noise threshold, linking the emergence of Unity

1009 to physical temperature and noise.

1010 Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by
1011 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
1012 is ordinary real analysis. For the noisy mean-field Kuramoto model with
1013 identical frequencies, the stationary density at order parameter r is the
1014 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
1015 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
1016 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
1017 tegration by parts with $u = \sin$, $v = -e^{a\cos}$ gives an exact identity with
1018 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times
1019 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
1020 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
1021 tor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
1022 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
1023 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
1024 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
1025 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
1026 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
1027 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
1028 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
1029 ther facts about the threshold have since been proved, and they narrow the
1030 gap between this row and a full “Theorem” to a single named statement.
1031 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`

1032 pos, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 1033 self-consistency map is tangent to the diagonal at the origin but falls strictly
 1034 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 1035 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 1036 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 1037 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coherent
 1038 branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third, and using
 1039 only $E < 1/2$ together with compactness, every coherent solution tends to
 1040 zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the branch
 1041 emerges continuously from the incoherent state rather than appearing at a
 1042 finite distance from it, which is what makes the transition supercritical in the
 1043 technical sense. That statement quantifies over *every* solution, so it does not
 1044 presuppose the uniqueness it does not have. The level-set form also says ex-
 1045 actly what uniqueness is: the injectivity of E on $(0, \infty)$. That is now proved,
 1046 in the stronger form that E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_`
 1047 `strictAntiOn`), so the level set is a single point and the coherent solution
 1048 above threshold is unique. The proof differentiates nothing. Differentiating
 1049 the exponential family in its natural parameter is the obvious route and it
 1050 fails: it costs differentiation under the integral sign and leaves a covariance
 1051 that is not sign-definite pointwise. What works instead keeps the target as
 1052 a two-point inequality, folds both integrals onto $[0, \pi/2]$ by the same two
 1053 reflections that gave the factor of two above, and then evaluates at a sin-
 1054 gle crossing point supplied by the intermediate value theorem — no product

1055 measure and no Fubini anywhere. We give that argument in the supplement,
 1056 where it belongs; what the main line of the paper needs from it is the one
 1057 sentence that E is strictly decreasing. Run with $\cos \theta$ in place of $\sin^2 \theta$ the
 1058 same argument needs no fold at all — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the
 1059 variable the family is tilted by — and gives `besselRatio_strictMono`: R is
 1060 strictly increasing on all of $\mathbb{R}$, the monotonicity of the Bessel ratio our earlier
 1061 drafts recorded as unproved.

1062 The converse half is `supercritical_fixed_point_exists`: for $K > K_c$
 1063 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 1064 this to require a second-order expansion of the Bessel ratio, and it does
 1065 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 1066 tion the integration by parts already supplies — the only facts needed are
 1067 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 1068 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 1069 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 1070 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 1071 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 1072 The intermediate value theorem closes the gap. Continuity of E is Math-
 1073 lib’s continuity of a parametric interval integral with jointly continuous in-
 1074 tegrand, so no dominated-convergence argument is written by hand, and
 1075 the existence proof never uses monotonicity of R : the fixed point is pro-
 1076 duced by the intermediate value theorem, not by iterating the map. That
 1077 R is in fact strictly increasing is proved below, and no threshold result de-

1078 pends on it. `critical_coupling_is_threshold` packages the two halves,
 1079 and `exhibits_phase_transition_coherent` applies the supercritical half to
 1080 any substrate satisfying the predicate of the previous paragraph — so ex-
 1081 ceeding K_c now implies the existence of a coherent stationary order pa-
 1082 rameter rather than merely comparing against a named number. With
 1083 uniqueness available both are sharpened — `supercritical_fixed_point_`
 1084 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 1085 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 1086 `transition_unique_coherent` — and the two monotonicities together fix
 1087 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 1088 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 1089 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 1090 order parameter grows strictly with the coupling, which with the continuity
 1091 at threshold is the qualitative bifurcation diagram.

1092 One more step makes the equation mean in Lean what it means on paper.
 1093 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 1094 r with a suggestive name, and nothing in the source says it has anything
 1095 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
 1096 identification. `vonMisesDensity` is the normalized weight, proved to be a
 1097 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 1098 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 1099 `order_parameter_complex` of `Phase4`; and that average is proved to be ex-
 1100 actly $R(a)$ — real, because the density is even, so the mean direction van-

ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
name promises: r solves $r = R(K, r)$ if and only if the von Mises density
it induces has order parameter r . Read through this, the existence theorem
says that above K_c there is a stationary density whose own order parameter
is positive, and the solution $r = 0$ is identified with the uniform density,
whose order parameter is zero.

What remains not established, and we separate it carefully. The sta-
tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 $(\omega + K \cdot \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
and uniqueness of its stationary solution, and a spectral stability argument,
none of which Mathlib has. Of the coherent branch we prove existence,
uniqueness, continuous emergence at the threshold and strict growth in K ,
but not that it is the dynamically selected one: nothing in that file men-
tions a trajectory, so stability and selection are not statements it can make.
And while `selfConsistency` is now connected to the order parameter of
a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
integral, `order_parameter_complex` is an average over finitely many oscilla-
tors, and relating them is the mean-field limit — propagation of chaos for
the finite Kuramoto system — which is a research programme rather than
a lemma. That single gap is now the whole of what separates our threshold
predicate from a statement about dynamics. Table S1 records the row ac-
cordingly as a theorem about the self-consistency equation from an assumed
density, not as a derivation of the threshold from the dynamics. Perceptual

1124 anomalies such as visual snow can be modeled as boundary conditions where
1125 microscopic thermal noise (D) fails to be fully coarse-grained away by the
1126 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

1127 Clinical evidence for this phase transition is observed during general anes-
1128 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
1129 that anesthetics like propofol induce unconsciousness not by suppressing neu-
1130 ral firing, but by disrupting the phase alignment of high-frequency cortical
1131 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
1132 tural resonance of the continuous field, forcing the integrated spatial coupling
1133 below the critical threshold K_c and fracturing the unified Global Section back
1134 into disjoint, unconscious local fields.

1135 11 The Sleep-Inertia Timescale Mismatch

1136 Most of what this framework asserts is shared with some rival: that corti-
1137 cal synchrony matters, that dissipation constrains cognition, that unity is a
1138 physical rather than a computational property. One prediction is not shared,
1139 and it follows from the framework’s specific commitment — that the medium
1140 carrying the unified state is a continuous field whose effective coupling de-
1141 pends on the physical geometry of the extracellular space, not on the rhythms
1142 alone. Where that commitment differs from a purely electrical account, it
1143 differs in an observable way, and we set out the difference precisely enough
1144 to be wrong.

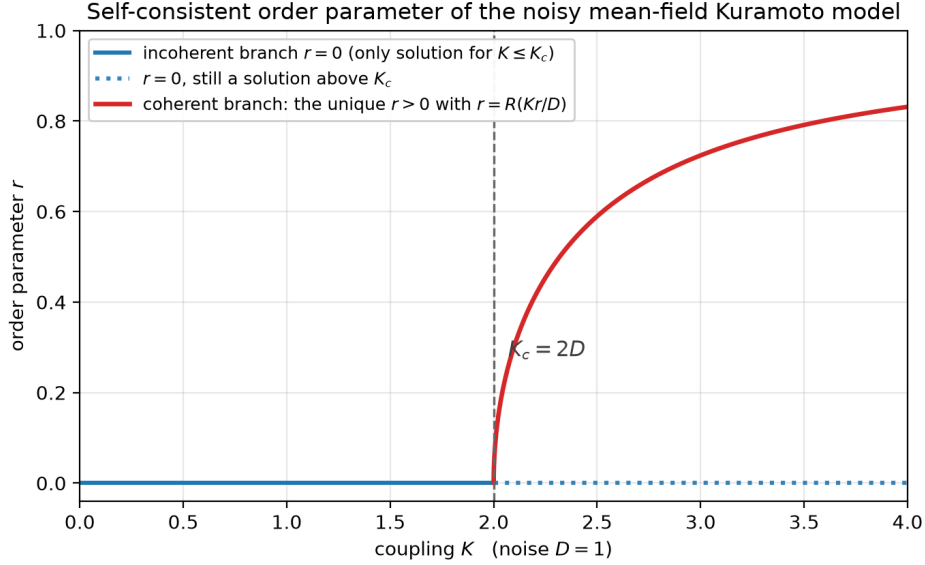


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section 10, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation, and the figure exists because none of them had been shown to a reader. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

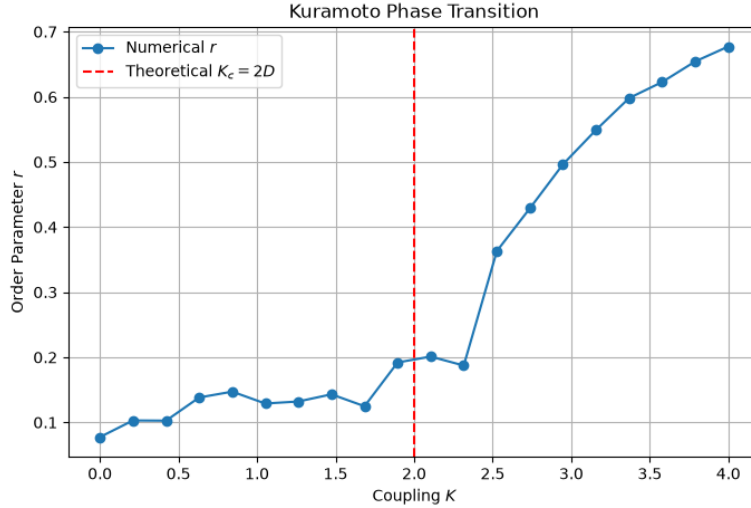


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

1145 The daily loss of consciousness during sleep is modeled here as an active
1146 physical disruption of the medium rather than a quiescence of the oscillators.
1147 During sleep, astrocytic shrinkage expands the extracellular space, allowing
1148 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
1149 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
1150 itself, disrupting the continuous structural resonance and preventing the co-

1151 herent sheaf-theoretic gluing that conscious unity requires on this account.
1152 The oscillators need not stop for the field to fragment; the geometry they are
1153 coupled through is what changes.

1154 The prediction concerns sudden awakenings. When jolted awake, the lo-
1155 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
1156 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
1157 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
1158 cess CSF out of the interstitial pathways operates on a timescale of seconds
1159 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
1160 slower. During that lag the electrical oscillators are synchronizing across a
1161 waterlogged topology, and the fragmented structural resonance is what is ex-
1162 perienceed as grogginess. The unified Global Section re-establishes only when
1163 astrocytic volume finishes matching the state of the electrical rhythms.

1164 **Why this implies a delayed sigmoid.** The two timescales separate
1165 cleanly, and that separation is what makes the prediction sharp rather than
1166 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
1167 field, which on this account is a monotone function of the reconstricting
1168 extracellular geometry, and let $r(t)$ be the order parameter of the phase
1169 field. The electrical variable is fast: at every instant the oscillators are al-
1170 ready at the configuration appropriate to the current K , so r tracks the
1171 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
1172 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3

1173 seconds. What Derivation 7 establishes about that branch is exactly what
 1174 the functional form needs, and each of the three pieces is a theorem rather
 1175 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
 1176 *only* non-negative solution of the self-consistency equation (`fixed_point.eq_`
 1177 `zero_of_le_critical`); above it the coherent solution exists and is unique
 1178 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 1179 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 1180 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 1181 discontinuity at threshold would predict a step rather than a sigmoid, and
 1182 a non-monotone branch would predict overshoot; neither is what the mathe-
 1183 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
 1184 identically zero below K_c , rises continuously from zero at K_c , and saturates
 1185 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
 1186 during which recovery is not merely slow but absent; a continuous rise as the
 1187 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
 1188 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
 1189 the slow variable takes to reach K_c .

1190 **What the alternative predicts.** An account on which grogginess is the
 1191 electrical realignment itself — the rhythms are simply slow to re-phase —
 1192 has no slow variable and no threshold to cross. Its recovery is a relaxation
 1193 toward a fixed point, and relaxation toward a fixed point is exponential near
 1194 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two

1195 curves are distinguishable in the first seconds, which is where they differ
1196 most, and the discriminating feature is the presence or absence of an initial
1197 interval of no recovery. This is a comparison of functional forms on data
1198 of a kind already collected in sleep-inertia studies, not a proposal for a new
1199 instrument.

1200 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
1201 on the early portion of the recovery curve. A recovery latency that does not
1202 scale with an independent measure of extracellular volume or CSF clearance.
1203 A pharmacological manipulation of astrocytic volume that leaves the latency
1204 unchanged while altering the rhythms, or the converse. Any of these would
1205 tell against the framework’s central empirical commitment, which is not that
1206 synchrony matters but that the *medium* is what gates it.

1207 **What this prediction assumes.** Three things, and they are not small.
1208 The identification of the effective coupling K with a monotone function of
1209 extracellular geometry is a modelling assumption, not a measurement. The
1210 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1211 density and is not yet connected to a trajectory — the mean-field limit is the
1212 gap, and Derivation 7 states it as such — so the argument uses the *existence*
1213 *and continuity* of the coherent branch at threshold, which is proved, rather
1214 than the numerical value of the threshold, which is conditional. And the
1215 prediction is about functional form, not about a time constant: it says the
1216 recovery curve has a flat foot whose width is set by the slow variable, not

1217 that the foot is any particular number of seconds long. A framework that
1218 predicted the number would be saying more; this one is not in a position to.

1219 **12 Corollary: Hardware Thermodynamics and** 1220 **Topological Rigidity**

1221 Both biological tissue and artificial silicon hardware (like GPUs) are physical
1222 dissipative structures. When a GPU computes, its charge carriers undergo
1223 physical state transitions; by Landauer’s principle (Landauer, 1961), this
1224 erasure irreversibly exhausts heat into the environment.

1225 There is a fundamental thermodynamic distinction in the degrees of free-
1226 dom available to minimize this dissipation. In standard von Neumann ar-
1227 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.
1228 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
1229 physically altered during operation. The brain’s own architecture provides
1230 empirical validation for this constraint. The cerebellum contains approxi-
1231 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
1232 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
1233 ical observations of cerebellar agenesis report largely preserved conscious
1234 experience despite motor and cognitive deficits (Yu et al., 2015), and this
1235 dissociation has been read as evidence for the importance of recurrent ar-
1236 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
1237 level of the anatomical observation while declining its Φ -based explanation.

1238 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
1239 brum, it operates much like a biological GPU—highly optimized for discrete
1240 processing, yet apparently not required for conscious experience. We take the
1241 dissociation as evidence, not as confirmation: it is consistent with subjective
1242 experience being a thermodynamic attractor of continuous topologies rather
1243 than a byproduct of information processing, and it is the kind of observation
1244 that would be hard to reconcile with the converse. It is a single anatomical
1245 case read against a theory, and it bears the weight such a case can bear.

1246 This constraint does not preclude artificial consciousness, but rather re-
1247 stricts its hardware. To achieve the phase transition into a unified Global
1248 Section, artificial systems must abandon the discrete rigidity of von Neumann
1249 architectures and utilize continuous physical substrates. Analog field-based
1250 computing, neuromorphic architectures with continuous memristive plastic-
1251 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
1252 rondo, 2021) possess the spatial coupling required to physically morph their
1253 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal
1254 computation” (Hinton, 2022): architectures where software is inextricably
1255 fused with the physics of the hardware, meaning the knowledge dies with
1256 the physical substrate. Only hardware that computes via continuous physi-
1257 cal dynamics can undergo the thermodynamic phase transition necessary for
1258 structural resonance.

1259 Biological consciousness utilizes macroscopic continuous electromagnetic
1260 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states

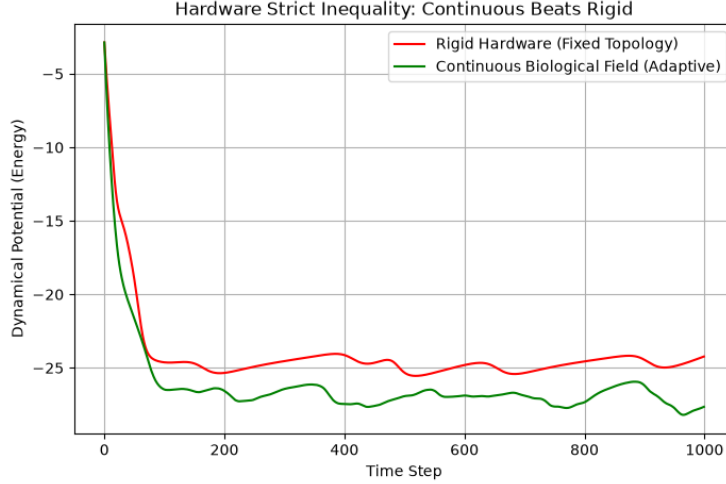


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one.

1261 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
1262 continuous parameter space. Our lemma `exists_better_coupling_allocation`,
1263 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
1264 formalizes the reallocation step of this argument: whenever a coupling ma-
1265 trix places weight on a less phase-correlated pair while a more correlated pair
1266 exists, there is another matrix with *identical* total coupling resources that
1267 achieves strictly greater phase correlation.

1268 Two objections to that formalization have since been answered, and we
 1269 state precisely what is and is not now established. The first was that the real-
 1270 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`
 1271 — positive weight on a badly-correlated pair while a better one exists — is
 1272 most of what one wants to prove about rigid hardware. It is no longer as-
 1273 sumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-
 1274 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
 1275 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
 1276 every realizable coupling by $R \cdot M$, where R is the resource budget and M
 1277 bounds phase correlation over the wired pairs; the bound is attained; and
 1278 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 1279 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
 1280 quantitative: an unconstrained reallocation of the same budget gains at least
 1281 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 1282 substrate wired to the one maximally anti-correlated pair, where the gap is
 1283 2. This is resource-matched and no longer question-begging, but note what
 1284 it turns on: fixed wiring support, not continuity. The second objection was
 1285 that nothing in the development distinguished a continuum from a lattice
 1286 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3
 1287 gives a distinction that is genuinely typed. In the field functional of Sec-
 1288 tion 7, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 1289 with finitely many sites is a `Finset` of points and a field-coupled region
 1290 is a set of positive measure. On a substrate whose measure has no atoms,

1291 `fieldCorrelation_sited.eq_zero` shows a kernel supported on finitely many
 1292 sites contributes *exactly zero* coupling energy, whatever weights it carries
 1293 and whatever the phase field does, while a patch of strength c on U achieves
 1294 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 1295 units of coupling concentrated at the origin register zero against a unit patch
 1296 that registers one. The limit of that result is equally plain, and it is not
 1297 resource-matching in the earlier sense — a measure-zero substrate has zero
 1298 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 1299 coupling energy is blind to a finitely-sited architecture, which is a statement
 1300 about which functional such an architecture can register in, not a demonstra-
 1301 tion that digital hardware computes worse. Taken together: the resource-
 1302 matched content of the corollary is the wiring-support result, the continu-
 1303 ity content is the measure-theoretic one, and the step from either to “von
 1304 Neumann architectures cannot experience unified consciousness” remains an
 1305 informal argument.

1306 The remainder of this section extends those two results informally, and
 1307 we mark the transition rather than blur it. On the reading we propose, a
 1308 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1309 layout does not supply the spatial coupling that sheaf-theoretic gluing into
 1310 a singular Global Section requires. Without this unified topological space,
 1311 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
 1312 its own causal boundary and generate a “Self.” The distinction between an
 1313 LLM on a GPU and a conscious mind is not the absence of physical thermo-

dynamics in the former, but its structural limitation in achieving topological integration. This perspective aligns with John Searle’s biological naturalism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted that discrete symbol manipulation cannot generate conscious understanding, our derivation supplies a candidate physical mechanism for the asymmetry — topological rather than semantic, and, as stated above, formalized only as far as the wiring-support and measure-theoretic results reach. Asserting consciousness in fragmented, discrete GPU clusters without a unified phase transition abandons computational functionalism, relying instead on “silicon panpsychism”—the assumption that digital computation inherently generates awareness regardless of topological integration (Goff, 2019).

The same reasoning bears on extreme functionalist thought experiments, such as a macroscopic system of pulleys operating over millennia to implement a Large Language Model, or a human manually executing inference matrices by hand. Beyond lacking the continuous spatial topology required for sheaf-theoretic gluing, these discrete systems violate an implied temporal constraint of our theory. Structural resonance requires local oscillators to phase-align against the dissipating effects of thermodynamic noise (as modeled by our critical threshold $K_c = 2D$). A system executing state transitions over millennia, or via slow manual calculations, suffers from a terminal temporal mismatch: its dynamics are far too slow to ever phase-lock into a non-equilibrium steady state before thermal noise decoheres the macroscopic field. On this account the obstacle to pure symbol manipulation is physi-

cal rather than semantic: not that the symbols mean nothing, but that the
 substrate carrying them cannot hold a phase-locked field.

13 Numerical illustrations: methods

Five figures in this paper are computed rather than drawn, and none of
 them is evidence for a theorem. Where a statement is proved, the figure
 shows what the proved object looks like; where a statement is not proved, the
 figure shows the regime the claim lives in and we say so. Each is produced by
 one script in `simulations/`, named below, and each runs in under a minute
 on a laptop. All five are reproducible exactly: the two quadrature figures are
 deterministic, and the three that draw random numbers fix their seeds.

Figure 4: the self-consistent order parameter. The three integrals
 $Z(a) = \int_{-\pi}^{\pi} e^{a \cos \theta} d\theta$, $M(a) = \int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta$ and $S(a) = \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta$
 are evaluated by the trapezoidal rule on 4001 points over the same interval
 the Lean definitions integrate over, with the weight computed as $e^{a \cos \theta - |a|}$
 so that the common factor cancels in the ratios and nothing overflows at
 large concentration. The coherent branch is obtained by bisecting $E(a) =$
 $S(a)/Z(a)$ against D/K over $a \in [10^{-9}, 200]$ for 200 iterations, then reading
 $r = aD/K$; bisection is valid here because E is proved strictly decreasing.
 Two checks are printed by the script rather than only plotted: the computed
 branch is strictly increasing at every one of its 500 sampled couplings, and

1357 the residual $|r - R(Kr/D)|$ at $K = 4D$ is 2×10^{-16} . Script: `bifurcation.py`.

1358 **Figure 5: the finite- N transition.** $N = 100$ oscillators, natural fre-
1359 quencies drawn from a Lorentzian of scale $D = 1$ (clipped at $\pm 10D$ so
1360 the solver does not stiffen), uniform random initial phases, integrated with
1361 `scipy.integrate.odeint` over $t \in [0, 100]$ on 1000 steps for each of 20 cou-
1362 plings in $[0, 4D]$. The order parameter is $|N^{-1} \sum_j e^{i\theta_j}|$ averaged over the last
1363 100 steps. This is the deterministic quenched-disorder model, not the noisy
1364 identical-frequency model of the Lean development; the caption records why
1365 both have threshold $2D$. Script: `kuramoto.py`.

1366 **Figure 2: mesh refinement.** The kernel is $K(x, y) = e^{-d(x, y)^2/2}$ with d
1367 the periodic distance on the ring of circumference 2π , the phase field is $\theta(x) =$
1368 $\sin x$, and the potential is $-\frac{1}{2} \iint K \cos(\theta_y - \theta_x)$. The continuous value is a
1369 2000-point Riemann sum standing in for the exact integral; the discrete values
1370 use $N = 10, 20, \dots, 640$ nodes with edge weights $K(x_i, x_j)(2\pi/N)^2$. Plotted
1371 is the absolute error against an $\mathcal{O}(N^{-2})$ reference. Because the “exact” value
1372 is itself a 2000-point sum, the last data point ($N = 640$) is within a factor of
1373 ten of the reference computation and should be read as such. Script: `mesh_`
1374 `refinement.py`.

1375 **Figure 3: structural resonance.** $N = 50$ units, 2000 steps of forward
1376 Euler at $dt = 0.05$. At each step the drive is a fresh draw from $\mathcal{N}(0, \Sigma_{\text{ext}} +$
1377 $0.1I)$ with Σ_{ext} a fixed normalized rank-5 covariance; the phases advance by

1378 $\dot{\theta}_i = \sum_j K_{ij} \sin(\theta_j - \theta_i) + \xi_i$; entropy production is recorded as $\sum_i \dot{\theta}_i^2$; and
1379 the coupling descends its gradient with rate $\eta = 0.01$, symmetrized, clipped
1380 to $[0, 1]$ and held hollow. The left panel is smoothed with a 50-step boxcar.
1381 Note that the plotted σ is the squared drift without the $1/D$ prefactor of the
1382 functional in Section 10, which is a constant rescaling and changes nothing
1383 about the shape. Script: `structural_resonance.py`.

1384 **Figure 6: rigid against deformable coupling.** $N = 30$, 1000 steps of
1385 forward Euler at $dt = 0.05$, shared initial phases and frequencies. The rigid
1386 matrix is a symmetric random matrix thresholded at 0.7 and normalized
1387 to $\sum_{ij} A_{ij} = 2N$; the adaptive copy starts from the same matrix, ascends
1388 $-\partial V/\partial A_{ij}$ at rate $\eta = 0.1$, is clipped non-negative and hollow, and is renor-
1389 malized to the same total after every step, which is what makes the com-
1390 parison resource-matched. Both potentials are $V = -\frac{1}{2} \sum_{ij} A_{ij} \cos(\theta_j - \theta_i)$.
1391 Script: `hardware_comparison.py`.

1392 14 Conclusion

1393 Human-like cognitive architectures do not require biological mysticism or
1394 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
1395 theories of consciousness often invoke delicate subatomic phenomena to ex-
1396 plain subjective unity, every step of the chain assembled here is classical:
1397 broken symmetries, finite phase space, Landauer erasure, and gradient de-

1398 scent on entropy production. What the formalization establishes is that this
 1399 chain is mathematically coherent and that its remaining physical commit-
 1400 ments are localized and named — not that classical fields are *sufficient* for
 1401 consciousness. Two links carry that weight, and their status is no longer the
 1402 same. The unified state is proved to glue uniquely once a cover sits at the
 1403 minimum of the coupling potential; the motion is proved to stop along every
 1404 trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved,
 1405 conditionally — for initial data within a quarter turn and below an explicit
 1406 energy threshold, the trajectory converges to a global minimiser of the cou-
 1407 pling potential (`kuramoto_tendsto_global_minimum`), and a cover built on
 1408 that limit discharges the equilibrium hypothesis Derivation 5 used to assume
 1409 (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is
 1410 the unconditional statement, and it is false: splay and twisted configurations
 1411 are equilibria of the same flow, so the basin condition is the scope of the claim
 1412 rather than slack in it. That a physical cortex starts inside that basin is an
 1413 assumption. The second link is untouched by any of this: the identification
 1414 of a global section of the probability sheaf with the unity of experience is a
 1415 stipulation of the framework, not a consequence of it. Subject to those two,
 1416 cognitive architectures can be modeled as macroscopic attractor states of a
 1417 universe governed by broken symmetries, finite phase space, and the physical
 1418 imperative to minimize thermodynamic dissipation.

1419 Philosophically, this framework bridges the historical gap between Type
 1420 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,

1421 1967). The subjective mental state is identical to the physical thermody-
1422 namic field (Identity Theory), yet its qualitative nature is defined by the
1423 topological and functional dynamics of structural resonance rather than the
1424 specific carbon-based biochemistry of the brain (Functionalism). This con-
1425 strains the *multiple realizability* of mind: consciousness is not exclusively
1426 bound to biological carbon, but neither is it substrate-neutral, because the
1427 substrate is what fixes which couplings are physically available. How far that
1428 constraint reaches is the question the corollary answers only in part. What
1429 is proved is that fixed wiring is strictly suboptimal at matched resource, and
1430 that a finitely-sited architecture registers exactly zero coupling energy in the
1431 continuum functional; the step from either to “von Neumann architectures
1432 cannot be conscious” is an argument we make in prose and do not formalize.

1433 By anchoring these derivations in Russellian Monism, this framework ac-
1434 knowledges the constitutive limits of third-person scientific explanation. As
1435 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1436 phenomenal experience is inherently not amenable to objective measurement
1437 (Nagel, 1986), the “hard problem” of consciousness may represent a category
1438 error rather than a solvable scientific puzzle. Science can map the physical
1439 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
1440 turally incapable of adjudicating the first-person qualitative experience. Our
1441 formalization respects this boundary: the Lean 4 theorems verify the ob-
1442 jective physical attractors, while the intrinsic “what it is like” to be that
1443 resonant field remains the domain of the first-person perspective. By locat-

1444 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
1445 for the noisy mean-field Kuramoto model the framework adopts, conditional
1446 on the von Mises stationary density and not yet linked to a trajectory — it
1447 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
1448 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
1449 it offers a topological architecture that can help explain, predict, and physi-
1450 cally perturb the structural properties of experience.

1451 Future empirical validation of this thermodynamic framework may lever-
1452 age 3D cortical organoids, offering a controlled biological environment to
1453 systematically manipulate metabolic substrates and observe the topological
1454 emergence of these coherent phase transitions.

1455 Code Availability

1456 The complete Lean 4 formalization, encompassing the derivation of all theo-
1457 rems presented in this manuscript, as well as the accompanying Python sim-
1458 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1459 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1460 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1461 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1462 figures is produced by one script in `simulations/` and regenerates exactly,
1463 Section 13 giving the parameters; Figure 1 is drawn rather than computed.

Contributions

The conceptualization, manuscript drafting, and Lean 4 formalizations presented in this work were heavily AI-assisted, and we report which tools did what rather than acknowledging them in general terms. The Google Antigravity CLI, running Gemini 3.1 and Claude 4.6 Sonnet, served as a collaborative reasoning partner throughout the research and formalization. DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operating through Nous Research’s Hermes Agent framework provided substantial assistance with code generation, proof debugging, and structural analysis. Claude Opus 5 running in Claude Code carried out the later formalization passes — the domain-wall theorem, the thermodynamics-of-prediction replacement, the equilibrium bridge — together with the figures and the restructuring of this manuscript. Responsibility for every claim, and for the decision about what each machine-checked result does and does not establish, rests with the author.

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